Math Appendix

Part A: Derivations of Equilibrium Conditions

Households: Setup, Nonhomothetic CES Aggregator

Following CLM (2017), the household CES aggregator over goods $j \in \mathcal{J}$ satisfies

$$\sum_{j \in \mathcal{J}} \left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}} \left(\frac{C_{i,t}^{j}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}} = 1, \tag{1}$$

with $\psi > 0$, $\mu_{i,t}^{j} > 0$, and $\sum_{j \in \mathcal{J}} e^{j} = 1$.

Households: Cost Minimization

The household solves

$$\min_{C_{i,t}^{j}} \sum_{j \in \mathcal{J}} P_{i,t}^{j} C_{i,t}^{j} \quad \text{s.t.} \quad (1). \tag{2}$$

FOC for $C_{i,t}^{j}$:

$$P_{i,t}^{j} = \lambda \left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}} \frac{\psi-1}{\psi} \left(\frac{C_{i,t}^{j}}{L_{i,t}}\right)^{-\frac{1}{\psi}} \frac{1}{L_{i,t}} \tag{3}$$

Relative demand ratio of two sectors (j and j'):

$$\frac{P_{i,t}^{j}}{P_{i,t}^{j'}} = \left(\frac{\mu_{i,t}^{j}}{\mu_{i,t}^{j'}}\right)^{\frac{1}{\psi}} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}(\varepsilon^{j}-\varepsilon^{j'})} \left(\frac{C_{i,t}^{j}}{C_{i,t}^{j'}}\right)^{-\frac{1}{\psi}} \tag{4}$$

Households: Demand Expressions

Solving (4) for $C_{i,t}^{j}$:

$$\left(C_{i,t}^{j}\right)^{\frac{1}{\psi}} = \left(\frac{\mu_{i,t}^{j}}{\mu_{i,t}^{j'}}\right)^{\frac{1}{\psi}} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}(\varepsilon^{j}-\varepsilon^{j'})} \left(\frac{P_{i,t}^{j'}}{P_{i,t}^{j}}\right) \left(C_{i,t}^{j'}\right)^{\frac{1}{\psi}} \tag{5}$$

$$\Rightarrow \quad \frac{C_{i,t}^{j}}{L_{i,t}} = \left(\frac{\mu_{i,t}^{j}}{\mu_{i,t}^{j'}}\right) \left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-\varepsilon^{j'})} \left(\frac{P_{i,t}^{j'}}{P_{i,t}^{j}}\right)^{\psi} \frac{C_{i,t}^{j'}}{L_{i,t}} \tag{6}$$

Plugging (6) *into the constraint* (1) *we have:*

$$\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}}\left(\left(\frac{\mu_{i,t}^{j}}{\mu_{i,t}^{j^0}}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-\varepsilon^{j^0})}\left(\frac{P_{i,t}^{j^0}}{P_{i,t}^{j}}\right)^{\psi}\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}=1$$

$$\left(\frac{1}{\mu_{i,t}^{j^0}}\right)^{\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)^2}{\psi}\varepsilon^{j^0}}\left(\frac{P_{i,t}^{j^0}}{1}\right)^{\psi-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}}\left(\left(\frac{\mu_{i,t}^{j}}{1}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)\varepsilon^{j}}\left(\frac{1}{P_{i,t}^{j}}\right)^{\psi}\right)^{\frac{\psi-1}{\psi}}=1$$

$$\left(\frac{1}{\mu_{i,t}^{j^0}}\right)^{\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)^2}{\psi}\varepsilon^{j^0}}\left(\frac{P_{i,t}^{j^0}}{1}\right)^{\psi-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}+\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}-\frac{(1-\psi)^2}{\psi}\varepsilon^{j}}\left(P_{i,t}^{j}\right)^{1-\psi}=1$$

$$\left(\frac{1}{\mu_{i,t}^{j^0}}\right)^{\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)^2}{\psi}\varepsilon^{j^0}}\left(\frac{P_{i,t}^{j^0}}{1}\right)^{\psi-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(\frac{1}{\psi}-\frac{(1-\psi)}{\psi})(1-\psi)\varepsilon^{j}}\left(P_{i,t}^{j}\right)^{1-\psi}=1$$

$$\left(\frac{1}{\mu_{i,t}^{j^0}}\right)^{\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)^2}{\psi}\varepsilon^{j^0}}\left(\frac{P_{i,t}^{j^0}}{1}\right)^{\psi-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)\varepsilon^{j}}\left(P_{i,t}^{j}\right)^{1-\psi}=1$$

$$\left(\frac{1}{\mu_{i,t}^{j^0}}\right)^{\frac{\psi-1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)^2}{\psi}\varepsilon^{j^0}+(1-\psi)}\left(\frac{P_{i,t}^{j^0}}{1}\right)^{\psi-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}}\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-1)}\left(P_{i,t}^{j}\right)^{1-\psi}=1$$

$$\left(\mu_{i,t}^{j^0}\right)^{\frac{1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{-(\psi-1)\varepsilon^{j^0}+\psi}{\psi}}\left(P_{i,t}^{j^0}\right)^{-1}\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{-\frac{1}{\psi}}\left[\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-1)}\left(P_{i,t}^{j}\right)^{1-\psi}\right]^{\frac{1}{1-\psi}}=1$$

Define the aggregate consumption price index, $P_{i,t}^{C}$, as follows:

$$P_{i,t}^{C}\equiv\left[\sum_{j\in\mathcal{J}}\left(\mu_{i,t}^{j}\right)\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-1)}\left(P_{i,t}^{j}\right)^{1-\psi}\right]^{\frac{1}{1-\psi}}. \tag{7}$$

Then,

$$\left(\mu_{i,t}^{j^0}\right)^{\frac{1}{\psi}}\frac{P_{i,t}^{C}}{P_{i,t}^{j^0}}=\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{1}{\psi}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(\psi-1)\varepsilon^{j^0}-\psi}{\psi}}$$

Solving for $\frac{C_{i,t}^{j^0}}{L_{i,t}}$ we have

$$\left(\frac{C_{i,t}^{j^0}}{L_{i,t}}\right)^{\frac{1}{\psi}}=\left(\mu_{i,t}^{j^0}\right)^{\frac{1}{\psi}}\frac{P_{i,t}^{C}}{P_{i,t}^{j^0}}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{(1-\psi)\varepsilon^{j^0}+\psi}{\psi}} \tag{8}$$

$$\frac{C_{i,t}^{j^0}}{L_{i,t}}=\mu_{i,t}^{j^0}\left(\frac{P_{i,t}^{C}}{P_{i,t}^{j^0}}\right)^{\psi}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)\varepsilon^{j^0}+\psi} \tag{9}$$

Households: Aggregate Price Index

Multiplying by prices ($P^j_{i,t}$) and summing over each sector we solve for total expenditure per worker:

$$\frac{1}{L_{i,t}}\sum_{j\in\mathcal{J}} P^j_{i,t}C^j_{i,t} = \sum_{j\in\mathcal{J}} \mu^j_{i,t}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j}\left(P^j_{i,t}\right)^{1-\psi}\left(P^C_{i,t}\right)^{\psi} \tag{10}$$

$$= \left(P^C_{i,t}\right)^{\psi}\cdot\frac{C_{i,t}}{L_{i,t}}\cdot\underbrace{\left[\sum_{j\in\mathcal{J}}\mu^j_{i,t}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j-1}\left(P^j_{i,t}\right)^{1-\psi}\right]}_{=\left(P^C_{i,t}\right)^{1-\psi}} \tag{11}$$

$$= \left(P^C_{i,t}\right)^{\psi}\cdot\frac{C_{i,t}}{L_{i,t}}\cdot\left(P^C_{i,t}\right)^{1-\psi} \tag{12}$$

$$= P^C_{i,t}\cdot\frac{C_{i,t}}{L_{i,t}} \tag{13}$$

Therefore $E^C_{i,t}\equiv\sum_j P^j_{i,t}C^j_{i,t} = P^C_{i,t}\,C_{i,t}$.

Households: Sectoral Final Consumption Expenditure Shares

Dividing sectoral expenditure by total we derive consumption expenditure shares: Multiply by sectoral price ($P^j_{i,t}$)

$$\frac{P^j_{i,t}\cdot\frac{C^j_{i,t}}{L_{i,t}}}{P^C_{i,t}\cdot\frac{C_{i,t}}{L_{i,t}}} = \mu^j_{i,t}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j}\left(P^j_{i,t}\right)^{-\psi}\left(P^C_{i,t}\right)^{\psi}\left(P^C_{i,t}\right)^{-1}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{-1} \tag{14}$$

$$\Rightarrow \quad s^j_{i,t} = \mu^j_{i,t}\left(\frac{P^j_{i,t}}{P^C_{i,t}}\right)^{1-\psi}\left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j-1} \tag{15}$$

Households: Investment CES Aggregator and Demand System

Investment has the same CES structure as consumption, but *homothetic*: set $\varepsilon^j = 1$ for all j in (1) and relabel $(\mu^j_{i,t},\psi)\to(\upsilon^j_i,\zeta)$:

$$\sum_{j\in\mathcal{J}}\left(\upsilon^j_i\right)^{\frac{1}{\zeta}}\left(\frac{X_{i,t}}{L_{i,t}}\right)^{\frac{1-\zeta}{\zeta}}\left(\frac{X^j_{i,t}}{L_{i,t}}\right)^{\frac{\zeta-1}{\zeta}} = 1. \tag{16}$$

Unlike the nonhomothetic case, (16) can be solved explicitly for the aggregate:

$$X_{i,t} = \left[\sum_{j\in\mathcal{J}}\left(\upsilon^j_i\right)^{\frac{1}{\zeta}}\left(X^j_{i,t}\right)^{\frac{\zeta-1}{\zeta}}\right]^{\frac{\zeta}{\zeta-1}}. \tag{17}$$

The household minimizes $\sum_j P^j_{i,t}X^j_{i,t}$ subject to (16).

FOC for $X^j_{i,t}$:

$$P^j_{i,t} = \lambda\left(\upsilon^j_i\right)^{\frac{1}{\zeta}}\left(\frac{X_{i,t}}{L_{i,t}}\right)^{\frac{1-\zeta}{\zeta}}\frac{\zeta-1}{\zeta}\left(\frac{X^j_{i,t}}{L_{i,t}}\right)^{-\frac{1}{\zeta}}\frac{1}{L_{i,t}}. \tag{18}$$

Dividing the FOCs for j and j':

$$\frac{P_{i,t}^{j}}{P_{i,t}^{j'}} = \left(\frac{\upsilon_i^j}{\upsilon_i^{j'}}\right)^{\frac{1}{\zeta}} \left(\frac{X_{i,t}^{j}}{X_{i,t}^{j'}}\right)^{-\frac{1}{\zeta}}. \tag{19}$$

The same steps as (6)–(9) (solve (19) for $X_{i,t}^{j}$, substitute into (16), solve), now with all income elasticities equal to one, give the sectoral demands and price index:

$$\frac{X_{i,t}^{j}}{L_{i,t}} = \upsilon_i^j \left(\frac{X_{i,t}}{L_{i,t}}\right) \left(P_{i,t}^{j}\right)^{-\zeta} \left(P_{i,t}^{X}\right)^{\zeta}, \qquad P_{i,t}^{X} \equiv \left[\sum_{j\in\mathcal{J}} \upsilon_i^j \left(P_{i,t}^{j}\right)^{1-\zeta}\right]^{\frac{1}{1-\zeta}}. \tag{20}$$

Multiplying (20) by $P_{i,t}^{j}$ and summing over j: $E_{i,t}^{X} \equiv \sum_j P_{i,t}^{j} X_{i,t}^{j} = P_{i,t}^{X} X_{i,t}$. Dividing (20) (times $P_{i,t}^{j}$) by $E_{i,t}^{X}$ gives the investment shares:

$$x_{i,t}^{j} \equiv \frac{P_{i,t}^{j} X_{i,t}^{j}}{P_{i,t}^{X} X_{i,t}} = \upsilon_i^j \left(\frac{P_{i,t}^{j}}{P_{i,t}^{X}}\right)^{1-\zeta}. \tag{21}$$

Firms: Unit Costs of Tradable Goods

Technology for tradable good $\omega^j \in [0,1]$:

$$q_{i,t}^{j}\left(\omega^j\right) = x_{i,t}^{j}\left(\omega^j\right) \left[k^{\varphi_i} l^{1-\varphi_i}\right]^{\beta_i^j} \left[\prod_{m\in\mathcal{J}} \left(D^{j,m}\right)^{\nu_i^{j,m}}\right]^{1-\beta_i^j}, \qquad \sum_{m\in\mathcal{J}} \nu_i^{j,m} = 1. \tag{22}$$

A producer with efficiency draw $x_{i,t}^{j}(\omega^j)$ has unit cost $c_{i,t}^{j}/x_{i,t}^{j}(\omega^j)$, where $c_{i,t}^{j}$ is the cost of one unit of the input bundle $F \equiv \left[k^{\varphi_i} l^{1-\varphi_i}\right]^{\beta_i^j} \left[\prod_m \left(D^{j,m}\right)^{\nu_i^{j,m}}\right]^{1-\beta_i^j}$:

$$c_{i,t}^{j} = \min_{k,\, l,\, \{D^{j,m}\}} r_{i,t}k + w_{i,t}l + \sum_{m\in\mathcal{J}} P_{i,t}^{m} D^{j,m} \quad \text{s.t.} \quad F = 1. \tag{23}$$

Each FOC, evaluated at $F = 1$, pins down spending on that input as a share of λ:
FOC for k:

$$r_{i,t} = \lambda\, \beta_i^j \varphi_i \frac{F}{k} \quad \Rightarrow \quad r_{i,t}k = \lambda\, \beta_i^j \varphi_i \tag{24}$$

FOC for l:

$$w_{i,t} = \lambda\, \beta_i^j (1-\varphi_i) \frac{F}{l} \quad \Rightarrow \quad w_{i,t}l = \lambda\, \beta_i^j (1-\varphi_i) \tag{25}$$

FOC for $D^{j,m}$:

$$P_{i,t}^{m} = \lambda\, (1-\beta_i^j) \nu_i^{j,m} \frac{F}{D^{j,m}} \quad \Rightarrow \quad P_{i,t}^{m} D^{j,m} = \lambda\, (1-\beta_i^j) \nu_i^{j,m} \tag{26}$$

Summing (24)–(26), total cost equals λ (the Cobb–Douglas shares add to one), so λ *is* the unit cost:

$$r_{i,t}k + w_{i,t}l + \sum_{m\in\mathcal{J}} P_{i,t}^{m} D^{j,m} = \lambda \Big[\beta_i^j \varphi_i + \beta_i^j (1-\varphi_i) + (1-\beta_i^j) \underbrace{\textstyle\sum_m \nu_i^{j,m}}_{=1}\Big] = \lambda \quad \Rightarrow \quad \lambda = c_{i,t}^{j}. \tag{27}$$

The input demands follow from the FOCs:

$$k = \frac{\beta_i^j \varphi_i \, c_{i,t}^j}{r_{i,t}}, \qquad l = \frac{\beta_i^j (1-\varphi_i) \, c_{i,t}^j}{w_{i,t}}, \qquad D^{j,m} = \frac{(1-\beta_i^j)\nu_i^{j,m} \, c_{i,t}^j}{P_{i,t}^m}. \tag{28}$$

Substituting (28) into $F = 1$:

$$\left[\left(\frac{\beta_i^j \varphi_i \, c_{i,t}^j}{r_{i,t}} \right)^{\varphi_i} \left(\frac{\beta_i^j (1-\varphi_i) \, c_{i,t}^j}{w_{i,t}} \right)^{1-\varphi_i} \right]^{\beta_i^j} \prod_{m \in \mathcal{J}} \left(\frac{(1-\beta_i^j)\nu_i^{j,m} \, c_{i,t}^j}{P_{i,t}^m} \right)^{(1-\beta_i^j)\nu_i^{j,m}} = 1. \tag{29}$$

The exponents on $c_{i,t}^j$ sum to one, so:

$$c_{i,t}^j = \kappa_i^j \left[\left((r_{i,t})^{\varphi_i} (w_{i,t})^{1-\varphi_i} \right)^{\beta_i^j} \left(\prod_{k \in \mathcal{J}} \left(P_{i,t}^k \right)^{\nu_i^{j,k}} \right)^{1-\beta_i^j} \right], \tag{30}$$

with constant

$$\kappa_i^j \equiv \left[\varphi_i^{\varphi_i} (1-\varphi_i)^{1-\varphi_i} \right]^{-\beta_i^j} \left(\beta_i^j \right)^{-\beta_i^j} \left(1-\beta_i^j \right)^{-(1-\beta_i^j)} \prod_{m \in \mathcal{J}} \left(\nu_i^{j,m} \right)^{-(1-\beta_i^j)\nu_i^{j,m}}. \tag{31}$$

Households: Dynamic Problem

The household chooses $\left\{ \bar{C}_{i,t}, B_{i,t+1} \right\}_{t=0}^{\infty}$ to maximize

$$\sum_{t=0}^{\infty} \delta^t L_{i,t} \, U\left(\bar{C}_{i,t} \right) \tag{32}$$

subject to the sequence of budget constraints

$$E_{i,t}^C \left(\bar{C}_{i,t}, \{ P_{i,t}^j \}_{j \in \mathcal{J}} \right) + P_{i,t}^X X_{i,t} + B_{i,t+1} + \tfrac{\phi_{i,t}}{2} \left(B_{i,t+1} - \bar{B}_i \right)^2 = w_{i,t} L_{i,t} + r_{i,t} K_{i,t} + R_{i,t} B_{i,t} + T_{i,t}. \tag{33}$$

Lagrangian:

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \delta^t L_{i,t} \, U\left(\bar{C}_{i,t} \right) + \sum_{t=0}^{\infty} \lambda_{i,t} \Big[& w_{i,t} L_{i,t} + r_{i,t} K_{i,t} + R_{i,t} B_{i,t} + T_{i,t} \\ & - E_{i,t}^C - P_{i,t}^X X_{i,t} - B_{i,t+1} - \tfrac{\phi_{i,t}}{2} \left(B_{i,t+1} - \bar{B}_i \right)^2 \Big]. \end{aligned} \tag{34}$$

Marginal consumption expenditure. From (7), $P_{i,t}^C$ depends on $\bar{C}_{i,t} \equiv C_{i,t}/L_{i,t}$. Differentiate both sides of (7) w.r.t. $\bar{C}_{i,t}$ and divide by $(1-\psi)$:

$$\left(P_{i,t}^C \right)^{-\psi} \frac{\partial P_{i,t}^C}{\partial \bar{C}_{i,t}} = \sum_{j \in \mathcal{J}} \mu_{i,t}^j \left(\varepsilon^j - 1 \right) \bar{C}_{i,t}^{(1-\psi)\left(\varepsilon^j - 1\right) - 1} \left(P_{i,t}^j \right)^{1-\psi}. \tag{35}$$

The shares (15) rearrange to $\mu_{i,t}^j \, \bar{C}_{i,t}^{(1-\psi)(\varepsilon^j - 1)} \left(P_{i,t}^j \right)^{1-\psi} = s_{i,t}^j \left(P_{i,t}^C \right)^{1-\psi}$, so each term in (35) equals $\left(\varepsilon^j - 1 \right) s_{i,t}^j \left(P_{i,t}^C \right)^{1-\psi} / \bar{C}_{i,t}$. Substituting, using $\sum_j s_{i,t}^j = 1$, and multiplying by $\left(P_{i,t}^C \right)^{\psi}$:

$$\frac{\partial P_{i,t}^C}{\partial \bar{C}_{i,t}} = \frac{P_{i,t}^C}{\bar{C}_{i,t}} \left(\sum_{j \in \mathcal{J}} \varepsilon^j s_{i,t}^j - 1 \right). \tag{36}$$

Product rule on $E^C_{i,t} = L_{i,t} P^C_{i,t} \bar{C}_{i,t}$, plugging in (36):

$$\frac{\partial E^C_{i,t}}{\partial \bar{C}_{i,t}} = L_{i,t}\left(P^C_{i,t} + \bar{C}_{i,t}\frac{\partial P^C_{i,t}}{\partial \bar{C}_{i,t}}\right) = L_{i,t}\, P^C_{i,t} \sum_{j\in\mathcal{J}} \varepsilon^j s^j_{i,t}. \tag{37}$$

FOC for $\bar{C}_{i,t}$, using (37) and $U\big(\bar{C}_{i,t}\big) = \ln \bar{C}_{i,t}$:

$$\delta^t L_{i,t}\, U'\big(\bar{C}_{i,t}\big) = \lambda_{i,t}\, L_{i,t}\, P^C_{i,t} \sum_{j\in\mathcal{J}} \varepsilon^j s^j_{i,t} \quad \Rightarrow \quad \lambda_{i,t} = \frac{\delta^t}{P^C_{i,t}\,\bar{C}_{i,t}\sum_{j\in\mathcal{J}} \varepsilon^j s^j_{i,t}}. \tag{38}$$

FOC for $B_{i,t+1}$ (appears in the budget constraint at t and at $t+1$):

$$\lambda_{i,t}\left[1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)\right] = \lambda_{i,t+1}\, R_{i,t+1} \quad \Rightarrow \quad \frac{\lambda_{i,t+1}}{\lambda_{i,t}} = \frac{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)}{R_{i,t+1}}. \tag{39}$$

Euler equation. Take the ratio of (38) at $t+1$ and t, set equal to (39), and rearrange:

$$\frac{P^C_{i,t+1}\,\bar{C}_{i,t+1}}{P^C_{i,t}\,\bar{C}_{i,t}} = \frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}} \cdot \frac{\delta\, R_{i,t+1}}{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)}, \qquad \widetilde{\phi}_{i,t} \equiv \left(\sum_{j\in\mathcal{J}} \varepsilon^j s^j_{i,t}\right)^{-1}. \tag{40}$$

My's attempt at an interpretation: $\frac{P^C_{i,t+1}\,\bar{C}_{i,t+1}}{P^C_{i,t}\,\bar{C}_{i,t}}$ can be interpreted as the growth rate of nominal consumption expenditure per worker. Therefore in equilibrium the optimal choice of Bonds to buy tomorrow will set this growth rate equal to the return on the bond ($R_{i,t+1}$) scaled by three terms, a nonhomothetic consumption wedge, intertemporal preference shifter, and financial friction.

$$\frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}}, \quad \frac{1}{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)}, \quad \text{and} \quad \delta. \tag{41}$$

The first term esecinally represents the growth rate of the marginal utility of a dollar caused purely by shifts in consumption towards services due to changes in expenditure weigted average income elasticities

$$\widetilde{\phi}_{i,t} \equiv \left(\sum_{j\in\mathcal{J}} \varepsilon^j s^j_{i,t}\right)^{-1}. \tag{42}$$

The second term represents an adjustment cost related to the distance a countries forgein asset position is from the specified baseline $\bar{B}$. Finally the third: some exogenous intertemporal preference shifter.

So all together if I understand it correctly:

$$\underbrace{\frac{P^C_{i,t+1}\,\bar{C}_{i,t+1}}{P^C_{i,t}\,\bar{C}_{i,t}}}_{\text{Expenditure Growth}} = \underbrace{\frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}}}_{\text{Structural Wedge}} \cdot \underbrace{\delta}_{\text{Exog. Parameter}} \cdot \underbrace{\frac{R_{i,t+1}}{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)}}_{\text{Effective Return}}$$

Households: Capital Accumulation Euler Equation

Add the law of motion to the Lagrangian with multiplier $\xi_{i,t}$:

$$\begin{aligned}\mathcal{L} = \sum_{t=0}^{\infty} \delta^t L_{i,t}\, U\big(\bar{C}_{i,t}\big) + \sum_{t=0}^{\infty} \lambda_{i,t}\Big[& w_{i,t}L_{i,t} + r_{i,t}K_{i,t} + R_{i,t}B_{i,t} + T_{i,t} - E^C_{i,t} - P^X_{i,t}X_{i,t} \\ & - B_{i,t+1} - \tfrac{\phi_{i,t}}{2}\left(B_{i,t+1} - \bar{B}_i\right)^2\Big] + \sum_{t=0}^{\infty} \xi_{i,t}\Big[(1-d)K_{i,t} + \chi_{i,t}\left(X_{i,t}\right)^{\sigma}\left(K_{i,t}\right)^{1-\sigma} - K_{i,t+1}\Big].\end{aligned} \tag{43}$$

FOC for $X_{i,t}$

$$\lambda_{i,t} P^X_{i,t} = \xi_{i,t}\, \sigma\, \chi_{i,t} \left(\frac{X_{i,t}}{K_{i,t}}\right)^{\sigma-1} \quad \Rightarrow \quad \xi_{i,t} = \frac{\lambda_{i,t}\, P^X_{i,t}}{\sigma\, \chi_{i,t}} \left(\frac{X_{i,t}}{K_{i,t}}\right)^{1-\sigma}. \tag{44}$$

FOC for $K_{i,t+1}$

$$\xi_{i,t} = \lambda_{i,t+1}\, r_{i,t+1} + \xi_{i,t+1} \left[(1-d) + (1-\sigma)\chi_{i,t+1} \left(\frac{X_{i,t+1}}{K_{i,t+1}}\right)^{\sigma}\right]. \tag{45}$$

Substitute (44) at t and t+1 into (45), divide by $\lambda_{i,t+1}$, and use the bond FOC $\lambda_{i,t}/\lambda_{i,t+1} = R_{i,t+1}/\left[1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)\right]$ from (39). Multiplying through by σ:

$$\frac{R_{i,t+1}}{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)} \frac{P^X_{i,t}}{\chi_{i,t}} \left(\frac{X_{i,t}}{K_{i,t}}\right)^{1-\sigma} = \sigma\, r_{i,t+1} + (1-\sigma)\frac{P^X_{i,t+1} X_{i,t+1}}{K_{i,t+1}} + (1-d)\frac{P^X_{i,t+1}}{\chi_{i,t+1}} \left(\frac{X_{i,t+1}}{K_{i,t+1}}\right)^{1-\sigma}. \tag{46}$$

Work through interpretation here:

Part B: Recovering the Model's Disturbances

Sectoral Productivities and Trade Costs

Inverting the price index $P^j_{i,t} = \Gamma\left[\Phi^j_{i,t}\right]^{-\frac{1}{\theta}}$:

$$\Phi^j_{i,t} = \Gamma^\theta \left(P^j_{i,t}\right)^{-\theta}. \tag{47}$$

Productivities. Set $h = i$ in the gravity equation (**??**) and use $\tau^j_{ii,t} = 1$ and (47):

$$\pi^j_{ii,t} = \frac{T^j_{i,t}\left(c^j_{i,t}\right)^{-\theta}}{\Phi^j_{i,t}} = \frac{T^j_{i,t}\left(c^j_{i,t}\right)^{-\theta}}{\Gamma^\theta\left(P^j_{i,t}\right)^{-\theta}} = T^j_{i,t}\left(\Gamma\,\frac{c^j_{i,t}}{P^j_{i,t}}\right)^{-\theta} \quad \text{for } j \in \mathcal{J}\backslash S \tag{48}$$

$$\Rightarrow \quad T^j_{i,t} = \pi^j_{ii,t}\left(\Gamma\,\frac{c^j_{i,t}}{P^j_{i,t}}\right)^{\theta}, \tag{49}$$

Trade costs. Divide (**??**) by the *exporter's* home share $\pi^j_{hh,t}$; the terms $T^j_{h,t}\big(c^j_{h,t}\big)^{-\theta}$ cancel:

$$\frac{\pi^j_{ih,t}}{\pi^j_{hh,t}} = \frac{T^j_{h,t}\left(c^j_{h,t}\tau^j_{ih,t}\right)^{-\theta}/\Phi^j_{i,t}}{T^j_{h,t}\left(c^j_{h,t}\right)^{-\theta}/\Phi^j_{h,t}} = \left(\tau^j_{ih,t}\right)^{-\theta}\frac{\Phi^j_{h,t}}{\Phi^j_{i,t}} = \left(\tau^j_{ih,t}\right)^{-\theta}\left(\frac{P^j_{i,t}}{P^j_{h,t}}\right)^{\theta}, \tag{50}$$

using (47) in the last step. Solving:

$$\tau^j_{ih,t} = \frac{P^j_{i,t}}{P^j_{h,t}}\left(\frac{\pi^j_{hh,t}}{\pi^j_{ih,t}}\right)^{\frac{1}{\theta}} \quad \text{for } j \in \mathcal{J}\backslash S. \tag{51}$$

Sectoral Demand Shifters

We recover $\mu^j_{i,t}$ by computing consumption expenditure two ways (supply and demand sides) and equating them.

Supply side. Multiply goods market clearing by $P^j_{i,t}$:

$$C^j_{i,t} + X^j_{i,t} + \sum_{m\in\mathcal{J}} D^{m,j}_{i,t} = Q^j_{i,t} \quad \Rightarrow \quad P^j_{i,t}C^j_{i,t} + P^j_{i,t}X^j_{i,t} + \sum_{m\in\mathcal{J}} P^j_{i,t}D^{m,j}_{i,t} = E^j_{i,t}, \tag{52}$$

where $E^j_{i,t} \equiv P^j_{i,t}Q^j_{i,t}$. Since $NX^j_{i,t} = Y^j_{i,t} - E^j_{i,t}$ with $Y^j_{i,t} = \sum_h \pi^j_{hi,t}E^j_{h,t}$, we have $E^j_{i,t} = Y^j_{i,t} - NX^j_{i,t}$. Substituting this and the investment shares (21), $P^j_{i,t}X^j_{i,t} = \upsilon^j_i \left(P^j_{i,t}/P^X_{i,t}\right)^{1-\zeta} P^X_{i,t}X_{i,t}$, isolates consumption expenditure:

$$P^j_{i,t}C^j_{i,t} = Y^j_{i,t} - NX^j_{i,t} - \sum_{m\in\mathcal{J}} P^j_{i,t}D^{m,j}_{i,t} - \upsilon^j_i \left(\frac{P^j_{i,t}}{P^X_{i,t}}\right)^{1-\zeta} P^X_{i,t}X_{i,t}. \tag{53}$$

Demand side. Multiply (9) by $P^j_{i,t}$:

$$\frac{P^j_{i,t}C^j_{i,t}}{L_{i,t}} = \mu^j_{i,t} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j} \left(P^j_{i,t}\right)^{1-\psi} \left(P^C_{i,t}\right)^{\psi}. \tag{54}$$

Since $E^C_{i,t} = P^C_{i,t}C_{i,t}$, $P^C_{i,t} = \left(E^C_{i,t}/L_{i,t}\right)/\left(C_{i,t}/L_{i,t}\right)$, so:

$$\left(P^C_{i,t}\right)^{\psi} = \left(\frac{E^C_{i,t}}{L_{i,t}}\right)^{\psi} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{-\psi} \quad \Rightarrow \quad \frac{P^j_{i,t}C^j_{i,t}}{L_{i,t}} = \mu^j_{i,t} \left(P^j_{i,t}\right)^{1-\psi} \left(\frac{E^C_{i,t}}{L_{i,t}}\right)^{\psi} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j-\psi}. \tag{55}$$

Equate and solve. Setting (53) equal to (55):

$$\mu^j_{i,t} = \frac{Y^j_{i,t} - NX^j_{i,t} - \sum_{m\in\mathcal{J}} \frac{(1-\beta^j_i)\nu^{j,m}_i}{P^j_i,_t} - \upsilon^j_i \left(\frac{P^j_{i,t}}{P^X_{i,t}}\right)^{1-\zeta} P^X_{i,t}X_{i,t}}{L_{i,t}\left(P^j_{i,t}\right)^{1-\psi} \left(\frac{E^C_{i,t}}{L_{i,t}}\right)^{\psi} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\varepsilon^j-\psi}} \quad \text{for } j \in \mathcal{J}\backslash s. \tag{56}$$

Financial Friction $\phi_{i,t}$

This friction is a maintained feature of the environment, not the object the counterfactual below is about: it lets the model match the US's actual, persistent trade imbalances instead of assuming balanced trade. The trade-cost counterfactual (Part D) is run both with it active and with it switched off, precisely to check that the trade-cost result does not depend on which of the two is true.

The Euler equation (40) contains two unknowns: the friction $\phi_{i,t}$ and the interest rate R_{t+1}. World bonds are in zero net supply, so there is a single world interest rate $R_{i,t+1} = R_{t+1}$, and bonds accumulate from net exports, $B_{i,t+1} = R_t B_{i,t} + NX_{i,t}$; the adjustment cost is rebated lump sum, so it does not enter the budget constraint net of itself.

First we rewrite the Euler equation in observables. Multiply (40) by $L_{i,t+1}/L_{i,t}$ so the left side becomes total expenditure growth, $E^C_{i,t} = L_{i,t}\, P^C_{i,t}\, \bar{C}_{i,t}$:

$$\frac{E^C_{i,t+1}}{E^C_{i,t}} = \frac{L_{i,t+1}}{L_{i,t}} \cdot \frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}} \cdot \frac{\delta\, R_{t+1}}{1 + \phi_{i,t}\left(B_{i,t+1} - \bar{B}_i\right)}. \tag{57}$$

This gives one equation per country: two equations, three unknowns ($\phi_{US,t}$, $\phi_{ROW,t}$, R_{t+1}).

We have to normalize (In the paper you normalized to unity, I could only make it work with 0 here)

$$\phi_{ROW,t} = 0. \tag{58}$$

With $\phi_{ROW,t} = 0$, the only unknown left in the ROW's equation is R_{t+1}:

$$R_{t+1} = \frac{1}{\delta} \cdot \frac{E^C_{ROW,t+1}}{E^C_{ROW,t}} \cdot \frac{L_{ROW,t}}{L_{ROW,t+1}} \cdot \frac{\widetilde{\phi}_{ROW,t}}{\widetilde{\phi}_{ROW,t+1}}. \tag{59}$$

With R_{t+1} known, the only unknown left in the US's equation is $\phi_{US,t}$:

$$\phi_{US,t} = \frac{1}{B_{US,t+1} - \bar{B}_{US}} \left[\frac{E^C_{US,t}}{E^C_{US,t+1}} \cdot \frac{L_{US,t+1}}{L_{US,t}} \cdot \frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}} \cdot \delta \, R_{t+1} - 1 \right]. \tag{60}$$

Part C: Solving the Model Forward

Part B takes prices as data; in the counterfactual they are unknowns, so the conditions of Part A have to be restated in changes. A prime is a counterfactual level and a hat the ratio of the two, $\widehat{x} \equiv x'/x$, taken within a year, so hatted objects suppress t. Working in ratios removes Γ, κ^j_i, $\mu^j_{i,t}$ and the levels of $T^j_{i,t}$ and $\tau^j_{ih,t}$, none of which then has to be measured.

Factor Markets: The Wage Is the Only Free Price

Total cost equals revenue, so the factor bills (24)–(25) scale to sector output by $Y^j_{i,t}$. Summing over sectors, with $VA_{i,t} \equiv \sum_j \beta^j_i Y^j_{i,t}$:

$$w_{i,t} L_{i,t} = (1 - \varphi_i) \, VA_{i,t}, \qquad r_{i,t} K_{i,t} = \varphi_i \, VA_{i,t}. \tag{61}$$

Both shares are fixed and both endowments are unchanged, $\widehat{K}_i = \widehat{L}_i = 1$, so

$$\widehat{w}_i = \widehat{r}_i = \widehat{VA}_i. \tag{62}$$

One number per country. We take it to be $\widehat{w}_i$, which is what enters the cost function; income follows as $VA'_i = VA_{i,t} \widehat{w}_i$.

Firms: Unit Costs in Changes

Taking (30) at the counterfactual over the baseline, κ^j_i cancels and by (62) the factor bundle collapses to $\widehat{w}_i$:

$$\widehat{c}^j_i = (\widehat{w}_i)^{\beta^j_i} \prod_{m \in \mathcal{J}} \left(\widehat{P}^m_i \right)^{\nu^{j,m}_i \left(1 - \beta^j_i\right)}. \tag{63}$$

Trade: Prices and Trade Shares in Changes

Prices. Since $P^j_{i,t} = \Gamma \left[\Phi^j_{i,t} \right]^{-1/\theta}$, in changes $\left(\widehat{P}^j_i \right)^{-\theta} = \Phi'^j_i / \Phi^j_i$ and Γ drops. Multiplying each term of the numerator by its own baseline counterpart over itself, the weight is the baseline trade share of (**??**):

$$\left(\widehat{P}^j_i \right)^{-\theta} = \frac{\sum_h T'^j_h \left(c'^j_h \tau'^j_{ih} \right)^{-\theta}}{\sum_k T^j_k \left(c^j_k \tau^j_{ik} \right)^{-\theta}} = \sum_h \underbrace{\frac{T^j_h \left(c^j_h \tau^j_{ih} \right)^{-\theta}}{\sum_k T^j_k \left(c^j_k \tau^j_{ik} \right)^{-\theta}}}_{= \pi^j_{ih,t}} \widehat{T}^j_h \left(\widehat{c}^j_h \widehat{\tau}^j_{ih} \right)^{-\theta}, \tag{64}$$

$$\left(\widehat{P}^j_i \right)^{-\theta} = \sum_h \pi^j_{ih,t} \, \widehat{T}^j_h \left(\widehat{c}^j_h \widehat{\tau}^j_{ih} \right)^{-\theta}. \tag{65}$$

Equations (63) and (65) are mutually dependent and solved together by iteration, which contracts at rate $1 - \beta^j_i$.

Trade shares. Taking (**??**) at the two equilibria, the denominator is $\Phi'^j_i / \Phi^j_i = \left(\widehat{P}^j_i \right)^{-\theta}$:

$$\pi'^j_{ih} = \pi^j_{ih,t} \, \widehat{T}^j_h \left(\frac{\widehat{c}^j_h \widehat{\tau}^j_{ih}}{\widehat{P}^j_i} \right)^{-\theta}. \tag{66}$$

Summing over h and using (65) returns one.

Trade in Levels: The Frozen Trade-Cost Counterfactual

Equations (65)–(66) take $\widehat{T}_h^j = \widehat{\tau}_{ih}^j = 1$ for granted: they compare the counterfactual year to *that same year's own* observed trade shares $\pi_{ih,t}^j$, so productivity and trade costs are implicitly held at whatever they actually were that year. Run across every year $t = 1, \ldots, T$ this reproduces the entire observed history of trade costs, which is exactly why it cannot be used to freeze trade costs: freezing τ at year t_0 while $t > t_0$ is not a change relative to year t's own baseline, so the trade share has to be rebuilt from levels instead of perturbed.

Levels, not changes. Let $T_{i,t}^j$ and $\tau_{ih,t}^j$ be the productivities and trade costs recovered in Part B, and $c_{i,t}^j$ the unit cost implied by the data at year t. The counterfactual unit cost is still given by (63), $c_i'^j = c_{i,t}^j \widehat{c}_i^j$, since factor markets and cost-minimization (Eqs. (62)–(63)) do not depend on trade costs. Define, for a chosen trade-cost tensor $\check{\tau}_{ih}^j$ (the data's own $\tau_{ih,t}^j$ to reproduce history, or τ_{ih,t_0}^j frozen at year t_0 for the counterfactual):

$$\Phi_i^j \equiv \sum_h T_{h,t}^j \left(c_{h,t}^j \, \tau_{ih,t}^j \right)^{-\theta}, \qquad \Phi_i'^j \equiv \sum_h T_{h,t}^j \left(c_h'^j \, \check{\tau}_{ih}^j \right)^{-\theta}. \tag{67}$$

Both sides of (47) use the same constant Γ, so

$$\widehat{P}_i^j = \left(\Phi_i'^j / \Phi_i^j \right)^{-1/\theta} \tag{68}$$

regardless of Γ's value, which is never needed numerically. Trade shares follow directly from the gravity equation (**??**) at the counterfactual:

$$\pi_{ih}'^j = \frac{T_{h,t}^j \left(c_h'^j \, \check{\tau}_{ih}^j \right)^{-\theta}}{\Phi_i'^j}. \tag{69}$$

Setting $\check{\tau} = \tau_{\cdot,t}$ (that year's own trade costs) in (67)– (69) must reproduce $\widehat{P}_i^j = 1$ and $\pi_{ih}'^j = \pi_{ih,t}^j$ to numerical precision, for every t in the sample – this is the check that T, τ and c recovered in Part B are mutually consistent, and it has to hold before the frozen-τ run ($\check{\tau} = \tau_{\cdot,t_0}$, every t) means anything. Steps 11–15 of the static block (real consumption, market clearing, the income residual) are unchanged; only $\widehat{P}$, $\widehat{c}$ and π' are rebuilt this way in place of (65)–(66).

Caveat on direction. $\tau_{ih,t}^j$ is recovered separately for each importer i, so the US-imports-from-ROW and ROW-imports-from-US directions can move independently. The ROW aggregate consolidates roughly two dozen countries, and its home trade share is close to frozen over the sample because intra-ROW trade counts as domestic absorption once consolidated – so the ROW-facing direction of frozen τ should be read as noisier than the US-facing direction.

Households: Budget Constraint and Investment

The adjustment cost is rebated, so those terms cancel in (33). Factor income is $VA_{i,t}$ by (61), and the bond terms collapse through $B_{i,t+1} - R_{i,t} B_{i,t} = NX_{i,t}$:

$$E_{i,t}^C = VA_{i,t} - E_{i,t}^X - NX_{i,t}. \tag{70}$$

The interest rate does not appear: the static block needs only net exports.

Investment. At $\sigma = 0$ the law of motion is $K_{i,t+1} = (1 - d + \chi_{i,t}) K_{i,t}$ and $X_{i,t}$ drops out, so (44) has a positive left side and a zero right side and the optimum is $X_{i,t} = 0$. Observed investment is positive, so we impose rather than derive $E_i^{X\prime} = x_{i,t} VA_i'$ with $x_{i,t} \equiv E_{i,t}^X / VA_{i,t}$. Then

$$E_i^{C\prime} = (1 - x_{i,t}) \, VA_{i,t} \widehat{w}_i - NX_{i,t}'. \tag{71}$$

Households: Inverting the Price Index

Equation (7) depends on the level of real consumption, so the consumption block cannot be taken in changes directly. Rearranging (15) term by term, $\mu_{i,t}^{j}\bar{C}_{i,t}^{(1-\psi)(\varepsilon^j-1)}\left(P_{i,t}^{j}\right)^{1-\psi} = s_{i,t}^{j}\left(P_{i,t}^{C}\right)^{1-\psi}$. Substituting that for each baseline term in the ratio of (7), with $\widehat{\bar{C}}_i = \widehat{C}_i$:

$$\left(\widehat{P}_i^{C}\right)^{1-\psi} = \sum_{j\in\mathcal{J}} s_{i,t}^{j}\left(\widehat{P}_i^{j}\right)^{1-\psi}\widehat{C}_i^{(1-\psi)(\varepsilon^j-1)}. \tag{72}$$

The shifter cancels, so nothing depends on the arbitrary normalization $\sum_j \mu_{i,t}^{j} = \kappa_i$. Since $\widehat{E}_i^{C} = \widehat{P}_i^{C}\widehat{C}_i$, substituting (72) leaves one equation in $\widehat{C}_i$:

$$\widehat{E}_i^{C} = \widehat{C}_i\left[\sum_{j\in\mathcal{J}} s_{i,t}^{j}\left(\widehat{P}_i^{j}\right)^{1-\psi}\widehat{C}_i^{(1-\psi)(\varepsilon^j-1)}\right]^{\frac{1}{1-\psi}}, \tag{73}$$

with the left side from (71). Differentiating its logarithm and using $\sum_j s_i'^{j} = 1$:

$$\frac{\partial\ln(\text{RHS})}{\partial\ln\widehat{C}_i} = 1 + \sum_{j\in\mathcal{J}} s_i'^{j}\left(\varepsilon^j - 1\right) = \sum_{j\in\mathcal{J}} s_i'^{j}\varepsilon^j = \frac{1}{\widetilde{\phi}_i}, \tag{74}$$

the reciprocal of the object in (40), so the root is unique.

Shares. Taking (15) at the two equilibria:

$$s_i'^{j} = s_{i,t}^{j}\left(\frac{\widehat{P}_i^{j}}{\widehat{P}_i^{C}}\right)^{1-\psi}\widehat{C}_i^{(1-\psi)(\varepsilon^j-1)}. \tag{75}$$

Summing over j reproduces (72). With equal income elasticities $\widehat{C}_i^{(1-\psi)(\varepsilon^j-1)}$ is common to all j and cancels against $\widehat{P}_i^{C}$: this term is the structural change mechanism.

Market Clearing: The Leontief System

Scaling (26) to sector output, the intermediate term in (52) is $\sum_m \nu_i^{m,j}\left(1-\beta_i^{m}\right)Y_{i,t}^{m}$, with gross output as the base and the buying sector m transposed relative to (22). Substituting it and eliminating Y with $Y_{i,t}^{j} = \sum_h \pi_{hi,t}^{j}E_{h,t}^{j}$ leaves the expenditures on both sides, but linearly:

$$E_i'^{j} = s_i'^{j}E_i^{C\prime} + \upsilon_i^{j}E_i^{X\prime} + \sum_{m\in\mathcal{J}} \nu_i^{m,j}\left(1-\beta_i^{m}\right)\sum_h \pi_{hi}'^{m}E_h'^{m}. \tag{76}$$

Stacking the country-sector pairs, with Π' mapping expenditure to output and Γ output back to input demand,

$$F_{(j,i)} = s_i'^{j}E_i^{C\prime} + \upsilon_i^{j}E_i^{X\prime}, \quad \Gamma_{(j,i),(m,i)} = \nu_i^{m,j}\left(1-\beta_i^{m}\right), \quad \Pi'_{(j,i),(j,h)} = \pi_{hi}'^{j}, \tag{77}$$

$$E' = \left[I - \Gamma\Pi'\right]^{-1}F, \qquad Y' = \Pi'E'. \tag{78}$$

Γ is block diagonal by country and holds no prices; Π' is block diagonal by sector and moves with the guess. The inverse exists because the column sums of $\Gamma\Pi'$ are bounded by the intermediate share:

$$\sum_{(j,i)}\left(\Gamma\Pi'\right)_{(j,i),(m,h)} = \sum_i \left(1-\beta_i^{m}\right)\pi_{hi}'^{m}\underbrace{\sum_j \nu_i^{m,j}}_{=1} \le \max_i\left(1-\beta_i^{m}\right) < 1. \tag{79}$$

Closing the Static Block

Output implies income through $VA'_i = \sum_j \beta_i^j Y_i'^j$, so a correct wage reproduces itself and the year is a scalar fixed point:

$$g\left(\widehat{w}_{US}\right) \equiv \frac{\sum_j \beta_{US}^j Y_{US}'^j}{VA_{US,t}\widehat{w}_{US}} - 1 = 0. \tag{80}$$

The residual falls in the guess: a higher wage raises costs by (63) and prices by (65), buyers shift away by (66), and implied income returns below the guess.

The redundant equation. Summing (52) over sectors returns (70), and over countries as well makes world output equal world expenditure identically, hence $\sum_i NX_{i,t} = 0$. One of the twenty-four static equations is implied by the others.

The numeraire. The system is homogeneous of degree one in prices, nominal quantities and NX, so it fixes only relative prices. Replacing the redundant equation with

$$\widehat{w}_R = 1 \tag{81}$$

and using $NX'_{R,t} = -NX'_{US,t}$ leaves $\widehat{w}_{US}$ as the only free number in the year.

Closing the Intertemporal Block

Equation (57) holds for $t = 1, \ldots, T-1$: $T-1$ restrictions on the T unknowns $\{NX'_{US,t}\}_{t=1}^{T}$. It restricts the tilt of the path, not its level, since shifting borrowing up in every year leaves every instance satisfied. Part A pins the level by transversality; truncating at T removes it, so we impose

$$B'_{US,T+1} = B_{US,T+1}. \tag{82}$$

Without it the counterfactual can borrow and never repay, and the comparison measures a windfall. It is a choice, and it binds at the last date, which is where the outcome is measured, so the answer's sensitivity to the horizon is a diagnostic.

The world interest rate is not an unknown: with $\phi_{ROW,t} = 0$ from Part B the rest of the world's saving equation contains only R_{t+1} and (59) gives it, now on the counterfactual consumption path.